AP PRECALCULUS

Unit 1 • Lesson 1.11 practice • Answer key

Lesson 1.11 • Equivalent Representations of Polynomial and Rational Expressions

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

Which is the standard form of $(x+1)^3$?

(A) x^3+1

(B) x^3+3x^2+3x+1

(C) x^3+3x+1

(D) x^3+2x^2+2x+1

Solution

Answer: (B). Row 3 of Pascal's triangle: $1, 3, 3, 1$.

Question 2

What is the remainder when x^3-2x+5 is divided by $x-2$?

(A) 1

(B) 5

(C) 9

(D) 13

Solution

Answer: (C). The remainder equals $p(2) = 8-4+5 = 9$.

Question 3

Which is the completely factored form of x^3-9x?

(A) $x(x-3)(x+3)$

(B) $x(x-3)^2$

(C) $(x-3)(x+3)$

(D) $x(x^2+9)$

Solution

Answer: (A). $x^3-9x = x(x^2-9) = x(x-3)(x+3)$.

Question 4

What is the coefficient of x^3 in the expansion of $(x+2)^5$?

(A) 10

(B) 20

(C) 80

(D) 40

Solution

Answer: (D). Row 5 is $1, 5, 10, 10, 5, 1$. The x^3 term is $10\,x^3 \cdot 2^2 = 40x^3$.

Question 5 (Free response)

Let $p(x) = 2x^3+x^2-13x+6$.

(a) Show that $x=2$ is a zero of p.

(b) Write $p(x)$ as a product of linear factors and find all zeros of p.

(c) Write $\dfrac{p(x)}{x+1}$ in the form $q(x)+\dfrac{k}{x+1}$.

Solution

(a) $p(2) = 16 + 4 - 26 + 6 = 0$.

(b) Synthetic division by 2 on $2, 1, -13, 6$ gives $2, 5, -3$ with remainder 0. So $p(x) = (x - 2)(2x^2 + 5x - 3) = (x - 2)(2x - 1)(x + 3)$. Zeros: 2, $\frac{1}{2}$, -3.

(c) Synthetic division by -1 gives $2, -1, -12$ with remainder 18: $\dfrac{p(x)}{x+1} = 2x^2 - x - 12 + \dfrac{18}{x+1}$.